FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

# JEE(Main)-2016

## ANSWERS, HINTS & SOLUTIONS

### FULL TEST – I
### (Main)

| S. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | A | B | A |
| 2. | C | B | C |
| 3. | A | B | C |
| 4. | C | C | B |
| 5. | D | A | A |
| 6. | A | B | B |
| 7. | A | C | C |
| 8. | A | A | D |
| 9. | D | B | A |
| 10. | B | A | A |
| 11. | B | C | B |
| 12. | B | B | A |
| 13. | D | C | C |
| 14. | B | A | B |
| 15. | D | C | A |
| 16. | B | C | C |
| 17. | A | C | C |
| 18. | B | A | B |
| 19. | C | C | C |
| 20. | A | A | D |
| 21. | C | B | A |
| 22. | A | D | C |
| 23. | B | B | C |
| 24. | C | A | A |
| 25. | C | C | B |
| 26. | B | C | D |
| 27. | A | A | B |
| 28. | A | A | A |
| 29. | C | B | A |
| 30. | B | A | D |

# Physics

## PART – I

### SECTION – A

1. For an adiabatic process $PV^{y} = k$
Here y = 3/2 and k = constant

$PV^{\frac{3}{2}} = k$

$\log P + \frac{3}{2}\log v = \log k$

$\frac{\Delta P}{P} + \frac{3}{2}\frac{\Delta v}{v} = 0$

$\frac{\Delta V}{V} = \frac{-2}{3}\frac{\Delta P}{P}$

$\frac{\Delta V}{V}\times 100 = \left(\frac{-2}{3}\right)\left(\frac{\Delta P}{P}\times 100\right) = \frac{-2}{3}\times\frac{2}{3} = \frac{-4}{9}$

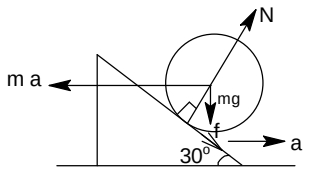
volume decreases by about $\frac{4}{9}\%$

2. $\tau_{\text{contact point}} = 0$

$\Rightarrow \tau_{mg} = \tau_{ma}$

$\Rightarrow R\, mg \sin 30^{o} = R\, ma \cos 30^{o}$

$\Rightarrow a = \frac{g}{\sqrt{3}}$



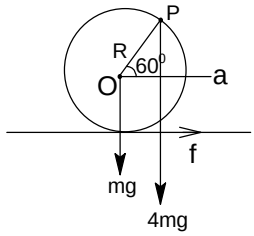

3. $\tau_{0} = I_{0}\alpha$

$\Rightarrow \frac{2}{5}mR\alpha + f = 2mg$

& f = ma = mαR

$\Rightarrow f = \frac{10}{7}mg \Rightarrow \frac{10}{7}mg \le \mu N$

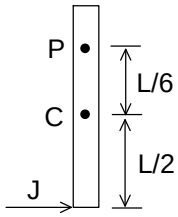
$\Rightarrow \frac{10}{7}mg \le \mu(5mg), \mu \ge \left(\frac{2}{7}\right)$




4. for ideal gas (i) and (iii) are correct
= for same pressure and no. of mole

$\frac{PV_1}{nT_1} < \frac{PV_2}{nT_2} \Rightarrow \frac{V_1}{T_1} < \frac{V_2}{T_2} \Rightarrow T_1 > T_2$

5. Conservation of linear momentum.




$J = MV_{f} \Rightarrow V_{f} = \frac{J}{M}$

Conservation of angular momentum about C

$$J\frac{L}{2}=\frac{ML^2}{12}\omega \Rightarrow \omega=\frac{6J}{ML}$$

$$\Rightarrow KE=\frac{1}{2}I_{cm}\omega^2+\frac{1}{2}MV_f^2=\frac{2J^2}{M}$$

6. Initially increase in temperature increases the density and after T = $4^oC$ increase in temperature decreases the density

7. Let $\ell$ is the length on potentiometer $i=\frac{5V}{5R}=1A$

$(1.5\ell)A=3 \Rightarrow$ † = 2m

8. $\because p=p_0\cos\left(\frac{2\pi}{\lambda}x\right)\Rightarrow\frac{p_0}{2}=p_0\cos\left(\frac{2\pi}{\lambda}\times1\right)$

$\Rightarrow \lambda = 6m \Rightarrow \nu = 60HZ$

9. $r=\sqrt{(8-2)^2+(-5-3)^2}=10$

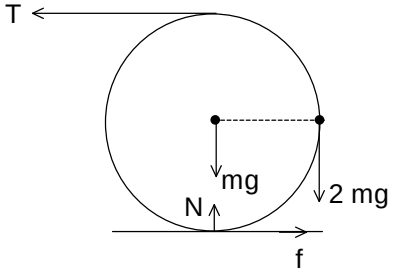
$$E=\frac{9\times10^9\times50\times10^{-6}}{100}=\frac{9}{2}\times10^3=4500\,V/m$$

10. $\because f'=\left(\frac{V_{sound}\pm V_0+V_{wind}}{V_{sound}\mp V_{source}+V_{wind}}\right)f$

$\therefore f' = f$

11. Place another identical hemisphere on given one, so that flat faces of two coincide. The total flux linked with cross- section of bottom half (lower hemisphere) is $\frac{Q}{2\epsilon_0}$ as charge enclosed by given hemisphere is zero. So flux linked with flat face of hemisphere is $\frac{-Q}{2\epsilon_0}$

12. $\because f = T$ …. (i)




N = mg + 2mg …. (ii)

(T + f)R = 2mg R …. (iii)

$\Rightarrow 2f = 2mg$

$\Rightarrow \mu N = mg$

$\Rightarrow \mu_{min}=\frac{1}{3}$

13. $\because \mu N = \mu m\omega^2 r = mg$

$\Rightarrow \mu = \dfrac{g}{\omega^2 r} = \dfrac{5}{\pi^2} = 0.506$

14. $\left[17 \times m\left({}_1H^1\right) + 18 \times m\left({}_0n^1\right) - M\right] \times 931 = 298$

15. $x = \left(\dfrac{\sqrt{7}-1}{3}\right)R_0$

16. Required kinetic energy = $\left[m(N) + m(n) - m(H) - m(C)\right] \times 931\,\text{MeV}$
= 2.99 MeV

17. $\because 10\beta' = \dfrac{11}{2}\beta$

$\Rightarrow 10\lambda'\dfrac{D}{d} = \dfrac{11}{2}\dfrac{\lambda D}{d}$

$\Rightarrow \dfrac{\lambda}{\lambda'} = \dfrac{20}{11} = \mu$

$\therefore \mu = 1.81$

18. Time constant will be 2RC

19. For refraction at bottom of slab

$\dfrac{4}{3V_1} - \dfrac{3}{2(45)} = 0 \Rightarrow V_1 = (+)40\,\text{cm}$

For mirror

$\dfrac{1}{-20} = \dfrac{1}{V_2} + \dfrac{1}{-60} \Rightarrow V_2 = (-)30\,\text{cm}$

For refraction at bottom of glass slab

$\dfrac{3}{2V_3} - \dfrac{4}{3(10)} = 0 \Rightarrow V_3 = +\dfrac{45}{4}\,\text{cm}$

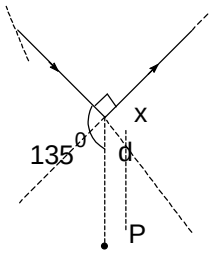
For refraction at upper surface of slab

$\dfrac{1}{V_4} - \dfrac{3/2}{-\left(45 - \dfrac{45}{4}\right)} = 0 \Rightarrow V_4 = (-)\dfrac{45}{2}\,\text{cm}$

20. $B = \dfrac{\mu_o I}{4\pi r}(\cos\alpha + \cos\beta) \times 2$

$\dfrac{\mu_o I}{2\pi\left(\dfrac{d}{\sqrt{2}}\right)}(\cos 0 + \cos 135)$

$= \dfrac{\mu_o I}{\sqrt{2}\pi d}\left(1 - \dfrac{1}{\sqrt{2}}\right)$




21. $\in = \dfrac{-d\phi}{dt}$

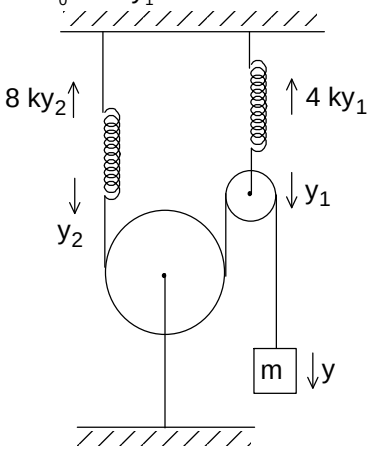
$$= \frac{d\left(\left(\pi R^2\right)\cdot B\right)}{dt}$$

$$\pi B \cdot 2R\frac{dR}{dt}$$

$2\pi B\left(R_o + t\right)\cdot 1$ anticlock wire

22. The current in wire AB, increase the flux through circular loop. Hence current in circular loop will be anticlockwise.

23. $\because 2F_0 = 4ky_1$




$F_0$ = Tension in the string

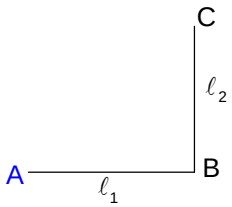
$F_0 = 8ky_2$

$$y = 2y_1 + y_2 = \frac{9F_0}{8K}$$

$$\Rightarrow K_{eq} = \frac{F_0}{y} = \left(\frac{8K}{9}\right)$$

$$\therefore T = 2\pi\sqrt{\frac{9m}{8K}}$$

24.




$V_x = V_x i + V_{yj}$

P.D = VBl

P.D = along y – axis = $VxBl_2$

P.D along x – axis = $V_yB.l_1$

P.D along AC = $V_xBl_2 + V_yBl_1$

25. Assume the hemisphere to be made of thin hemispherical layers.

26. Co-efficient of mutual inductance = $\mu_0 n_1 n_2 \ell \pi r_1^2$

$= \mu_0 \pi r_1^2$

Hence $M = \dfrac{\mu_0 \pi r^2}{2R}$

27. $\because 200 \times g = x^2 \times 2 \times 1 \times g$

$\Rightarrow x = 10\,cm$

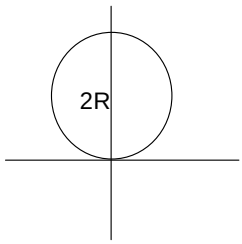
$\therefore V = x^3 = 1000\,cm^3$

28. Electric field and magnetite field will be parallel and velocity will also parallel to electric filed

$\Rightarrow$ All three are parallel

Hence only electric force will act on it

29. $y = 2R = 2 \times \dfrac{mv_o}{qB} = \dfrac{2v_o}{\alpha \cdot B_0}$




30. $\because V_T = \dfrac{2r^2 g}{9\eta}(d - \rho)$

$\Rightarrow \dfrac{2 \times 10^{-2}}{3600} = \dfrac{2r^2 \times 10}{9 \times 10^{-2} \times 10^{-1}}\left(1.8 \times 10^3 - 10^3\right)$

$\Rightarrow r \approx 1.77 \times 10^{-6}\,m$

$\therefore D = 2r = 3.54 \times 10^{-6}\,m$

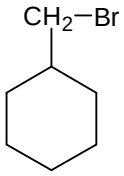
# Chemistry

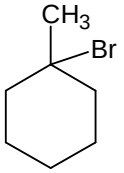
## PART – II

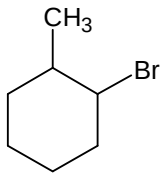
### SECTION – A

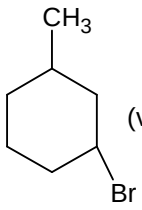
1. $K_2Cr_2O_7 + 14HCl \rightarrow 2KCl + 2CrCl_3 + 3Cl_2 + 7H_2O$

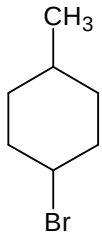
   n-factor of HCl = $\frac{6}{14} = \frac{3}{7}$

2. (i) 

(ii) 

(iii) 

(iv) 

(v) 


3. $PV = nRT \Rightarrow P = nRT / V$

4. $PV^{\gamma} = \text{const} \Rightarrow P\gamma V^{\gamma-1}dV + V^{\gamma}dP = 0$

   $\frac{dP}{P} = -\gamma \cdot \frac{dV}{V}$

5. $\Delta G^{o} = -0.059 \log k = -nFE^{0}_{cell} \therefore k < 1, \Delta G > 0$

6. $\left[Br^{-}\right]_{CaBr_2} = 0.01 \times 2, \quad \left[Br^{-}\right]_{KBr} = 0.01 \quad \left[Ag^{+}\right]_{AgNO_3} = 0.05$

   Both $Br^-$ & $Ag^+$ ions act as common ions, so larger the conc. of $Br^-$ or $Ag^+$ more is the suppression of ionization of gBr

7. Because shielding effect and increase in effective nuclear charge are cancelled

8. Not lies in same plane

9. NOF has lone pair while $NO_2F$ has no lone pair

10. $Zn + NaOH(A) \rightarrow Na_2ZnO_2 + H_2(B)$

    $NaOH + P_4(C) \rightarrow NaH_2PO_2 + PH_3$

    $NaOH + NH_4Cl \rightarrow NaCl + NH_3(D) + H_2O$

11. Due to low electronegativity of N in compound C basic character decreases. Due to umbrella effect, A is least reactive.

12. Non super imposable mirror Image

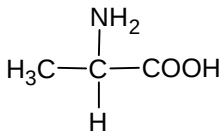
13. Due to cyclopropyl unique stability.

14. In (II) negative charge present on O atom while positive on C. (I) is most stable due to neutral.

15. Each ring contributes 1DBE.

16. Ring expansion will take place

17. Aniline is



18. Facts.

19. $\left.\begin{array}{l} NaCl + H_2SO_4 \xrightarrow{\Delta} NaHSO_4 + HCl \\ NaCl + NaHSO_4 \xrightarrow{\Delta} Na_2SO_4 + HCl \end{array}\right\}$ – Craft process

20. Bond strength $\alpha$ thermal stability

21. Fact.

22. $(X)\underset{\text{green}}{2Cr^{3+}} + 10OH^- + 3H_2O_2 \rightarrow \underset{\text{yellow}}{2CrO_4^{2-}}$

$(Y)CrO_4^{2-} + 2H_2O_2 + 2H^+ \rightarrow \underset{[\text{blue}]}{2CrO_5} + 3H_2O \rightarrow Cr_2O_3 + 7/2O_2 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + 3H_2O$

23. Fact

24. Fact

25. (cis/trans) i.e. Geometrical

26. Due to strong ligand $t_{2g}^6\, e_g^o$

27. Self reduction
$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$

28. $P = K_H\, x$
$\log P = \log K_H + \log X$

29. Electraode potential is intensive property while gibb's free energy is extensive

30.

| | | | |
|---|---|---|---|
| $A$ | $\longrightarrow$ | $2B + C$ | |
| $P_i$ | | $0 \quad 0$ | Pressure at $t = 0$ |
| $P_i - P$ | | $2P \quad P$ | Pressure at any t |

$\therefore P_t = P_i + 2P$

or $P = \dfrac{P_t - P_i}{2}$

$\because K = \dfrac{2.303}{t}\log\left(\dfrac{P_i}{P_i - P}\right)$

$= \dfrac{2.303}{t}\log\left(\dfrac{P_i}{P_i - \dfrac{P_t - P_i}{2}}\right) = \dfrac{2.303}{t}\log\left(\dfrac{2P_i}{3P_i - P_t}\right)$

# Mathematics

## PART – III

### SECTION – A

1. $0 \le \frac{|x|}{1+x^2} < 1 \therefore m = \left[\frac{|x|}{1+x^2}\right] = 0$

$1 \le 2 - \sin 3x \le 3$

$\frac{1}{3} \le \frac{1}{2-\sin 3x} \le 1$

$\therefore n = \text{integral value of } \frac{1}{2-\sin 3x} = 1$

$\therefore m \ne n$

2. at the point $\sin x = \frac{\sqrt{3}}{2}$

$P = \left(\frac{2\pi}{3}, \frac{3}{2}\right)$

$\frac{3}{2} = \frac{\sqrt{3}}{2}.\frac{2\pi}{3} + a$

$a = \left(\frac{3}{2} - \frac{\pi}{\sqrt{3}}\right) \quad a \in \left(\frac{3}{2} - \frac{\pi}{\sqrt{3}}, \infty\right)$

3. $f_P(\alpha) = e^{\frac{i\alpha}{P^2}(1+2+\ldots P)} = e^{\frac{i\alpha}{P^2}\frac{P(P+1)}{2}} = e^{\frac{i\alpha}{2}\left(1+\frac{1}{P}\right)}$

$\therefore \lim_{n\to\infty} f_n(\pi) = \lim_{n\to\infty} e^{\frac{i\pi}{2}\left(1+\frac{1}{n}\right)} = i$

4. $\frac{z-1}{i} = \text{real}, \ \frac{x+iy-1}{i} = \text{real}$

$\frac{x-1}{i} + y = \text{real} \quad \therefore x-1=0, x=1$

$\sin^{-1}\left(\frac{z-1}{i}\right) = \sin^{-1}(y) \Rightarrow -1 \le y \le 1$

$\text{Re}|z| = x = 1, -1 < \text{Im}(z) \le 1$

5. $S = \sum_{r=1}^{16} \frac{8r}{(4r^4+1)}$

$\sum_{r=1}^{16} \frac{8r}{(2r^2-2r+1)(2r^2+2r+1)} = 2\sum_{r=1}^{16}\left(\frac{1}{(2r^2-2r+1)} - \frac{1}{2r^2+2r+1}\right)$

$= 2\left(\frac{1}{1} - \frac{1}{5} + \frac{1}{5} - \frac{1}{13} + \frac{1}{13} + \ldots \frac{1}{481} - \frac{1}{545}\right)$

$= 2\left(1 - \frac{1}{545}\right) = \frac{1088}{545}$

6. 13, $a_1$, $a_2$ …. $A_{20}$, 67 are in AP

$\therefore a_1 + a_2 .... + a_{20} = 20\left(\frac{13+67}{2}\right) = 800$

AM $\geq$ GM

$\therefore \frac{a_1 + a_2 + a_3 .... + a_{20}}{20} \leq (40)^{20}$

Maximum value is $(40)^{20}$.

7. $\sqrt{2}\left(\cos\left(\cos x.\cos\frac{\pi}{4} + \sin(\cos x)\left(\sec\frac{\pi}{4}\right)\right)\right)$

$= \sqrt{2}\left\{\cos\left(\cos x - \frac{\pi}{4}\right)\right\}$

$= -1 \leq \cos\left(\cos x - \frac{\pi}{4}\right) \leq 1$

$-\sqrt{2} \leq A \leq \sqrt{2}$

8. $\frac{99\pi}{2} \leq \theta \leq 50\pi$

$\frac{3\pi}{2} \leq \theta \leq 2\pi$

$x = -\sin^2\theta$, $\cos y = \cos^2\theta$

$y - x = 1$

9. $\sqrt{2}\left(\frac{1}{\sqrt{2}}\cos\frac{\pi}{2n} + \frac{1}{\sqrt{2}}\sin\left(\frac{\pi}{2n}\right)\right) = \frac{\sqrt{n}}{2}$

$\Rightarrow \sqrt{2}\cos\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) = \frac{\sqrt{n}}{2}$

$\cos\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) = \frac{\sqrt{n}}{2\sqrt{2}}$

$n = 6,\ \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

10. $\sin x > 0$ & $\sin 3x > 0$

$\log_2(2\sin x \sin 3x) \geq 0$

$\Rightarrow 2\sin x.\sin 3x \geq 1$

$3x \in (0,\pi) \cup (2\pi, 3\pi)$

$x \in \left(0, \frac{\pi}{3}\right) \cup \left(\frac{2\pi}{3}, \pi\right)$

for $2\sin x \sin 3x \geq 1$

$2\sin^2 x(3 - 4\sin^2 x) \geq 1$

$8\sin^4 x - 6\sin^2 x + 1 \leq 0$

$\Rightarrow (2\sin^2 x - 1)(4\sin^2 x - 1) \leq 0$

$$\Rightarrow \left(\sin x - \frac{1}{\sqrt{2}}\right)\left(\sin x + \frac{1}{\sqrt{2}}\right)\left(\sin x - \frac{1}{2}\right)\left(\sin x + \frac{1}{2}\right) \le 0$$

$$\frac{1}{2} \le \sin x \le \frac{1}{\sqrt{2}}$$

$$x \in \left[\frac{\pi}{3}, \frac{\pi}{4}\right] \cup \left[\frac{2\pi}{3}, \frac{3\pi}{4}\right]$$

finally $x \in \left(\frac{2\pi}{3}, \frac{3\pi}{4}\right]$

11. Required no = $5!\left(\frac{1}{2!} - \frac{1}{3!} - \frac{1}{4!} - \frac{1}{5!}\right)$

$$= 120\left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120}\right)$$

$$= 60 - 20 + 5 - 1 = 44$$

12. $\left(1 + x + x^3 + x^4\right)^n = \left[\left(1+x\right)\left(1+x^3\right)\right]^n$

$$= \left(1+x\right)^n \left(1+x^3\right)^n$$

$$= \left(1 + {}^nC_1 x + {}^nC_2 x^2 \ldots + {}^nC_1 x^n\right)\left(1 + {}^nC_1 x^3 + {}^nC_2 x^6 + \ldots . {}^nC_n x^{3n}\right)$$

Coefficient of $x^{15}$

$$= \left({}^nC_5 \times {}^nC_0\right) + \left({}^nC_4 \times {}^nC_3\right) + \left({}^nC_3 \times {}^nC_6\right) + \ldots + \left({}^nC_2 \times {}^nC_9\right) + \left({}^nC_3 \times {}^nC_{12}\right) + \left({}^nC_6 \times {}^nC_{15}\right)$$

$$= \sum_{r=0}^{5} {}^nC_{5-r}\, {}^nC_{3r}$$

13. $P\left(\frac{B}{A \cup \bar{B}}\right) = \frac{P\left(B \cap \left(A \cup \bar{B}\right)\right)}{P\left(A \cup \bar{B}\right)} = \frac{P\left(A \cap B\right)}{P\left(A \cup \bar{B}\right)}$

Now $P\left(A \cap \bar{B}\right) = PA - P\left(A \cap B\right)$

$$\Rightarrow P\left(A \cap B\right) = P\left(A\right) - P\left(A \cap \bar{B}\right)$$

$$= \frac{7}{10} - \frac{1}{2} = \frac{1}{15}$$

Also $P\left(A \cup \bar{B}\right) = P\left(A\right) + P\left(\bar{B}\right) - P\left(A \cap \bar{B}\right)$

$= \frac{7}{10} + \frac{3}{5} - \frac{1}{2} = \frac{4}{5}$, $P\left(\frac{B}{A \cup \bar{B}}\right) = \frac{1}{4}$

14. Putting $x = 0$, $a_0 = 1$

Differentiate both side and putting $x = 0$ we get $a_1 = 4a$

Differentiate again and putting $x = 0$

$a_2 = 6a^2 + 4b$

$\Delta = -\left(a_0^3 + a_1^3 + a_2^3 - 3a_0 a_1 a_2\right) \quad \because \; a_0 + a_1 + a_2 \neq 0$

$1 = 4a = 6a^2 + 4b$

$a = \frac{1}{4}$, $b = \frac{5}{32}$

15. $a_{11} = 0, a_{12} = 3, a_{21} = 3, a_{22} = 0$

$A = \begin{vmatrix} 0 & 3 \\ 3 & 0 \end{vmatrix} \quad |A| = 9$

$A^{-1} = \frac{-1}{9}\begin{bmatrix} 0 & -3 \\ -3 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{3} & 0 \end{bmatrix}$

16. $CD = \frac{16+5+181}{\sqrt{4+25}} = \sqrt{29}$

AD = 3

$\Rightarrow CA^2 = 9 + 29 = 38$

Equation of circle is $(x-3)^2 + (y+1)^2 = 38$

17 shortest distance will take place along the common normal

Equation of normal for $y^2 = 4ax$ at $(at^2, 2at)$

$y = -tx + t + t^3$

If it passes through (6, 0) then

$t^3 - 5t = 0$

$t = 0, t = \pm\sqrt{5}$

$\Rightarrow A(5, 2\sqrt{5}), C = (5, -2\sqrt{5})$

$PA = PC = \sqrt{1+20} = \sqrt{21}; OP = 6$

Minimum distance = $(\sqrt{21} - \sqrt{5})$ units

18. Let the mid point of chord be P(h, K); its equation will be

$xh - yK = h^2 - K^2$

$y = \frac{h}{K}x + \frac{K^2 - h^2}{K}$

It should be same as $y = mx + \frac{a}{m}$

$\Rightarrow m = \frac{h}{K}$ and $\frac{a}{m} = \frac{K^2 - h^2}{K}$

$\Rightarrow a = \frac{K^2 - h^2}{K}\frac{h}{K} \Rightarrow K^2 a = K^2 h.h^3$

$\Rightarrow h^3 = K^2(h - a)$

Locus is $x^3 = y^2(x - a)$

19. equation of tangents is $SS_1 = T^2$

$\left(\frac{x^2}{9} + \frac{y^2}{4} - 1\right)\left(\frac{\lambda^2}{9} + \frac{9}{4} - 1\right) = \left(\frac{\lambda x}{9} + \frac{3y}{4} - 1\right)^2$

Right angle a + b = 0

Solve it get $\lambda = \pm 2$

20. $\lim\limits_{x\to 0}\dfrac{x^a\left(\dfrac{\sin x}{x}\right)^b}{\sin\left(\dfrac{x^c}{x^c}\right).x^c} \Rightarrow \lim\limits_{x\to 0} x^{a+b-c}\dfrac{\left(\dfrac{\sin x}{x}\right)^b}{\left(\dfrac{\sin x^c}{x^c}\right)}$

The value of limit is non-zero only

$a + b - c = 0$

21. $\lim\limits_{x\to 1}\int\limits_4^{f(x)}\dfrac{2tdt}{x-1}$

$\lim\limits_{x\to 1}\dfrac{2f(x)\times f'(x)}{1} = 2f(1).f'(1)$

$= 8f'(1)$

22 $y = f(x)$

$x = f^{-1}(y)$

$f(x) = x + \tan x$

$y = f^{-1}(y) + \tan^{-1} f^{-1}(y)$

$y = g(y) + \tan(g(y))$

$x = g(x) + \tan(g(x))$

Differentiate

$1 = g'(x) + \sec^2 g(x).g'(x)$

$g'(x) = \dfrac{1}{1+\sec^2(g(x))} = \dfrac{1}{2+(g(x)-x)^2}$

23. $f(x) = \cos|x| - 2ax + b$

$= \cos x - 2ax + b$

$f'(x) = -\sin x - 2a$

$f'(x) \geq 0$

$-\sin x - 2a \geq 0$

$a \leq -\dfrac{1}{2}$

24. Let $x^x = t$

$x\log x = \log t$

$(\log x + 1)dx = \dfrac{1}{t}dt$

$t(1+\log x)dx = dt$

$I = \int\dfrac{1}{t\sqrt{t^2-1}}dt = \sec^{-1}(x^x) + c$

25. $\lim\limits_{n\to\infty} 2^K\left(\dfrac{1^K + 2^K + 3^K + ...n^K}{n^{K+1}}\right)$

$$= 2^K \lim_{n\to\infty} \sum \frac{1}{n}\left(\frac{r}{n}\right)^K$$

$$2^K \int_0^1 x^K dx = 2^K \left[\frac{x^{K+1}}{K+1}\right]_0^1 = \frac{2K}{K+1}$$

26. $I = 2\int_1^4 \frac{e^{\sqrt{x}}}{2\sqrt{x}}\sqrt{x}\,dx$

$= 2\int_1^2 ze^z dz \qquad z = \sqrt{z}$

$= 2\left[ze^z - e^z\right]_1^2 = 2\left[2e^2 - e^2 - 0\right] = 2e^2$

27. Area $= \int_0^{\frac{\pi}{4}} \sin y\,dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos y\,dy$

$= -\left[\cos y\right]_0^{\frac{\pi}{4}} + \left[\sin y\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$= \frac{-1}{\sqrt{2}} + 1 + 1 - \frac{1}{\sqrt{2}} = \left(2 - \sqrt{2}\right)$ sq unit

28. $x\cos y\frac{dy}{dx} + \sin y = x^2.e^x$

$\Rightarrow x\cos y dy + \sin y dx = x^2 e^x dx$

$d\left(x.\sin y\right) = x^2 e^x dx$

$x\sin y = x^2 e^x - 2\int x.e^x dx$

$= x^2 e^x - 2xe^x + 2e^x + c$

$x\sin y = e^x\left(x^2 - 2x + 2\right) + c$

29. $\left(\vec{a}\times\vec{b}\right).\left(\left(\vec{a}\times\vec{c}\right)\times\vec{d}\right)$

$= \left(\vec{a}\times\vec{b}\right).\left(\left(\vec{a}.\vec{d}\right)\vec{c} - \left(\vec{c}.\vec{d}\right)\vec{a}\right)$

$= \left(\vec{a}.\vec{d}\right)\left[\vec{a}\vec{b}\vec{c}\right] - \left(\vec{c}.\vec{d}\right)\left[\vec{a}\vec{b}\vec{c}\right]$

$= \left(\vec{a}.\vec{d}\right)\left[\vec{a}\vec{b}\vec{c}\right]$

30. let the plane $ax + by + cz = 1$ & here $\frac{x}{6} = \frac{y}{4} = \frac{z}{3}$

When $6a + 4b + 3c = 0$

$2a - b = 1$

$3a - 4b + 5c = 1$

$a = \frac{29}{85}, b = \frac{-27}{85}, c = \frac{-22}{85}$

$29x - 27y - 22z = 85$